Equivalent Expressions: Distribute, Combine, Factor
Answer Key

Grade 6–7 Mathematics

Quick Answers

1. $4x + 12$
2. (a) $= 3\,(x + 5)$, (b) $= 3x + 15$, —
3. $9x - 5$
4. $3\,(2x + 5)$
5. $-8x + 14$
6. $4x + 3$ cm
7. (a) $= 4\,(s + 2)$, (b) $= 4s + 8$
8. $3\,(3x + 8)$
9. $2x + 16$
10. $8x + 23$

What is verified

9 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problem 2. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 6–7 Mathematics
7.EE.A – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*
Difficulty 1–4 of 5 across 13 tagged checks.

1. Use the distributive property to expand: $4(x + 3)$

Solution: Multiply the 4 by *each* term inside: $4 \cdot x + 4 \cdot 3$. $\boxed{4x + 12}$

2. A rectangular garden bed is 3 m deep, with width split into x m of tomatoes and 5 m of peppers. (a) area as a product; (b) area as a sum; (c) explain why they are equal.

Solution: (a) The whole width is $(x + 5)$ m and the depth is 3 m, so the area is depth times width: $\boxed{3(x + 5)}$

(b) Distribute the 3 over both width pieces: $3 \cdot x + 3 \cdot 5$. $\boxed{3x + 15}$

(c) Model answer: the tomato section has area $3x$ and the pepper section has area 15 square meters, and together those two pieces tile exactly the same region the product $3(x+5)$ measures, so the two expressions must be equal. (Any one-sentence answer that identifies both section areas with the whole area earns credit — instructor judges the sentence.)

3. Combine like terms: $7x + 4 + 2x - 9$

Solution: Group the like terms: $(7x + 2x) + (4 - 9)$. The x-terms give $9x$; the constants give -5. $\boxed{9x - 5}$

4. Factor out the greatest common factor: $6x + 15$

Solution: The GCF of 6 and 15 is 3. Write each term as 3 times something: $6x = 3 \cdot 2x$ and $15 = 3 \cdot 5$. $\boxed{3(2x + 5)}$

5. Expand: $-2(4x - 7)$

Solution: Distribute -2 to *both* terms, watching the signs: $-2 \cdot 4x = -8x$ and $-2 \cdot (-7) = +14$. $\boxed{-8x + 14}$

6. The sides of a triangle measure x cm, $(x + 4)$ cm, and $(2x - 1)$ cm. Write and simplify an expression for the perimeter.

Solution: Perimeter is the sum of the three sides: $x + (x + 4) + (2x - 1)$. Combine the x-terms ($x + x + 2x = 4x$) and the constants ($4 - 1 = 3$). $\boxed{4x + 3 \text{ cm}}$

7. 4 friends each buy a sandwich costing s dollars and a juice costing 2 dollars. (a) total cost as a product; (b) as a sum.

Solution: (a) Each friend spends $(s + 2)$ dollars and there are 4 friends: $\boxed{4(s + 2)}$

(b) Distribute the 4: $4 \cdot s + 4 \cdot 2$. $\boxed{4s + 8}$

8. Factor out the greatest common factor: $9x + 24$

Solution: The GCF of 9 and 24 is 3: $9x = 3 \cdot 3x$ and $24 = 3 \cdot 8$. $\boxed{3(3x + 8)}$

9. Simplify completely: $3(2x + 5) - 4x + 1$

Solution: Distribute first: $3(2x + 5) = 6x + 15$. Then combine like terms: $6x + 15 - 4x + 1 = (6x - 4x) + (15 + 1)$. $\boxed{2x + 16}$

10. **Challenge.** Simplify completely: $5(2x + 3) - 2(x - 4)$

Solution: Distribute both products, keeping the signs: $5(2x + 3) = 10x + 15$ and $-2(x - 4) = -2x + 8$. Combine: $(10x - 2x) + (15 + 8)$. $\boxed{8x + 23}$